

Akarshika Tripathi

Seattle, WA 98030 (open to relocate) | (928)221-1811 | akatripathi2000@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Software Engineer with 4+ years of experience building scalable web, mobile, and distributed applications using JavaScript, React.js, Node.js, Python, and Java. Skilled in designing APIs, SDKs, and backend microservices, with hands-on expertise in AWS, Docker, and CI/CD pipelines. Experienced in data modeling, recommendations, and integrating machine learning solutions (TensorFlow, PyTorch, LangChain) into production systems. Proven ability to deliver reliable, high-performance applications that enhance customer experience, system scalability, and business impact.

EXPERIENCE

Northern Arizona University | Graduate Research Assistant

May 2024 – December 2025

Working on the Cyclist Routing Algorithm for Network Connectivity (CRANC), a tool designed to enhance urban bicycling safety and accessibility by providing tailored routes based on cyclist preferences.

- Designed and implemented the [CRANC](#) cyclist routing system using TypeScript, Python, improving routing accuracy by 20%.
- Built full-stack modules (React.js, Java, Flutter) and optimized UI frameworks, enhancing responsiveness by 25%.
- Deployed applications on Docker, reducing deployment cycles by 40% and ensuring system scalability and reliability.
- Applied data modeling with OpenStreetMap datasets to enhance real-time route planning and mapping accuracy by 20%.
- Collaborated with cross-disciplinary researchers and engineers to ensure robust, distributed, and customer-focused system design.

Larix Technologies | Software Developer

April 2021 – December 2023

- Developed and scaled Chatbot.team, an omnichannel SaaS platform serving thousands of users, enhancing customer engagement and business efficiency.
- Built backend microservices (Node.js, MySQL) with event-driven and distributed architectures, improving system efficiency by 40%.
- Created and integrated REST APIs across platforms, boosting automation and reducing manual intervention by 40%.
- Built reusable React.js UI components and optimized API design, cutting feature delivery time by 25%.
- Directed technical design, SDK components, and prototypes for customer integrations, reducing design-to-delivery cycles by 25%.
- Facilitated agile ceremonies (sprint planning, backlog grooming, client reviews), increasing throughput by 30% while maintaining high coding standard.
- Partnered with product managers and engineers to align system architecture, scalability, and reliability with business goals.
- Implemented CI/CD pipelines and automated testing frameworks, improving release reliability and reducing deployment errors by 30%.

PROJECTS

DocuGenie – AI-powered Document QA System

[Live Demo](#)

- Built with Node.js (Express), React, AWS EC2, and Vercel to enable conversational querying of PDFs.
- Integrated LangChain, embeddings, and LLMs (vector search, caching) to deliver reliable, scalable ML-powered document insights.
- Applied metadata handling and semantic understanding for streamlined user experience.

AIBI – Business Intelligence Agent

[Live Demo](#)

- Developed in Python + AWS EC2 to analyze structured data (Google Sheets, CSVs, Stripe) with natural language queries.
- Applied ML-powered reasoning for KPI summaries, forecasts, and recommendations.
- Delivered a low-cost, scalable BI solution enabling non-technical teams to make data-driven decisions.

SKILLS

- Languages: JavaScript, TypeScript, Python, Java, Dart, SQL, Familiar with Golang
- Frameworks & Tools: React.js, Node.js, Knex.js, Vue.js, Flutter, Android Studio, Bootstrap
- Databases: MySQL, PostgreSQL
- DevOps & Cloud: AWS (EC2, Lambda, S3, ECS, DynamoDB, Redis, SQS), Azure, Docker
- Core Concepts: Distributed Systems, Concurrency, API Design, Data Modeling, SDK Development, Recommendations, Search, Scalability, Reliability, ML Integration (TensorFlow, PyTorch, LangChain)
- Collaboration Tools: Git, GitHub, Bitbucket, Postman, Jira, Figma, Scrum

EDUCATION

Northern Arizona University

December 2025

- Master of Science in Information Technology (GPA: 4.0)
- Awards:** Engineering FEST 4 Winner – 1st Place for developing Safe & Secure Peer-to-Peer Marketplace
 - Relevant Coursework:** Web Technology, Databases, Blockchains, Applied Cryptography, Applied Business Intelligence, Data Warehousing & Business Analytics, Information Security Concepts & Administration

CERTIFICATIONS

- Information Security Management Fundamentals
- Google Analytics Certification
- Crash Course on Python by Google
- Blockchain Certification Training

LEADERSHIP & IMPACT

- Entrepreneur Lead – NSF I-Corps Hub (Desert & Pacific Region).
- 3x Star Employee of the Month – Larix Technologies.
- Volunteer – Hope Foundation NGO (Hunger Drives, Animal Rescue)